Ratio, Root, and Alternating Series Tests

Section 1: The Ratio Test

1 Evaluate the limit $L = \lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right|$ for the series $\sum_{n=1}^{\infty} \frac{3^n}{n!}$.

a. 0 (converges)

b. 3 (diverges)

c. 1 (inconclusive)

d. ∞ (diverges)

2 When is the Ratio Test inconclusive?

a. When $L = 1$

b. When $L < 1$

c. When $L > 1$

d. When $L = 0$

3 Determine the convergence of $\sum_{n=1}^{\infty} \frac{n!}{2^n}$.

a. Diverges because $L = \infty > 1$

b. Converges because $L = 0 < 1$

c. Inconclusive ($L = 1$)

d. Converges by Alternating Series Test

Section 2: The Alternating Series Test

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4 What two conditions must an alternating series $\sum (-1)^{n-1} b_n$ with $b_n > 0$ satisfy to converge by the AST?

a. $b_{n+1} \le b_n$ for all n, and $\lim_{n \to \infty} b_n = 0$

b. b_n is increasing, and $\lim_{n \to \infty} b_n = 1$

c. $\lim_{n \to \infty} b_n = 0$ only

d. The Ratio Test limit must be < 1

5 Does the alternating harmonic series $\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n}$ converge or diverge?

a. Converges conditionally

b. Converges absolutely

c. Diverges

d. Oscillates without limit

6 Determine the convergence of $\sum_{n=1}^{\infty} \frac{(-1)^n}{\sqrt{n}}$.

a. Converges conditionally

b. Converges absolutely

c. Diverges

d. Inconclusive

Section 3: Absolute vs. Conditional Convergence

7 A series $\sum a_n$ is said to converge absolutely if which condition holds?

a. $\sum |a_n|$ converges

b. $\sum a_n$ converges but $\sum |a_n|$ diverges

c. $\lim_{n \to \infty} a_n = 0$

d. All terms are positive

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8 Which of the following series converges absolutely?

a. $\sum_{n=1}^{\infty} \frac{(-1)^n}{n^2}$

b. $\sum_{n=1}^{\infty} \frac{(-1)^n}{n}$

c. $\sum_{n=1}^{\infty} \frac{(-1)^n}{\sqrt{n}}$

d. $\sum_{n=1}^{\infty} (-1)^n$

9 If a series converges absolutely, MUST it also converge in its original form?

a. Yes, absolute convergence implies convergence

b. No, it may oscillate

c. Only if it is alternating

d. Only for geometric series

Section 4: Alternating Series Error Bound

10 For a convergent alternating series $\sum (-1)^{n-1} b_n$, what is the upper bound for the error $|R_n| = |S - S_n|$?

a. $|R_n| \leq b_{n+1}$

b. $|R_n| \leq b_n$

c. $|R_n| \leq \frac{b_{n+1}}{2}$

d. $|R_n| \leq \frac{b_n}{n}$

11 If the sum of $\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n^3}$ is approximated by its first 4 terms, what is the maximum possible error?

a. $\frac{1}{125}$

b. $\frac{1}{64}$

c. $\frac{1}{216}$

d. $\frac{1}{25}$

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12 How many terms of the alternating series $\sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n}$ are needed to guarantee an error of less than 0.01?

a. 100 terms

b. 99 terms

c. 10 terms

d. 50 terms

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